HcZZ ctag2d -- Results, no Z+X (2026-09-23)

Card: combine_run3_100150_ctag2d_rawsf_nozx_jecshifts_automcstats_2026_09_23
hplusc_mva_4class_ctag2d_scored_v8 + hplusc_mva_4class_ctag2d_jecshifts_scored_v2,
[100,150] GeV, MC only, 20 qqZZ-quantile bins, 23 named nuisances +
autoMCStats(10). CMS_ctag2d: official up_Total/down_Total, unmodified (L0 swap correction
reverted since 2026-09-15). Other_Higgs lhe_pdf/scalevar_muR/muF shapes included since
2026-09-20 (only the private HPlusBottom sub-sample masked). 61.9 fb^-1, 13.6 TeV.

Yields [100,150] GeV

Process	Yield
Signal	0.053130
ggZZ	6.677889
qqZZ	36.318223
Other_Higgs	55.077609
Total bkg	**98.073721**

$S/\sqrt{B}$ = 0.0054

Expected limits (AsymptoticLimits, --run blind)

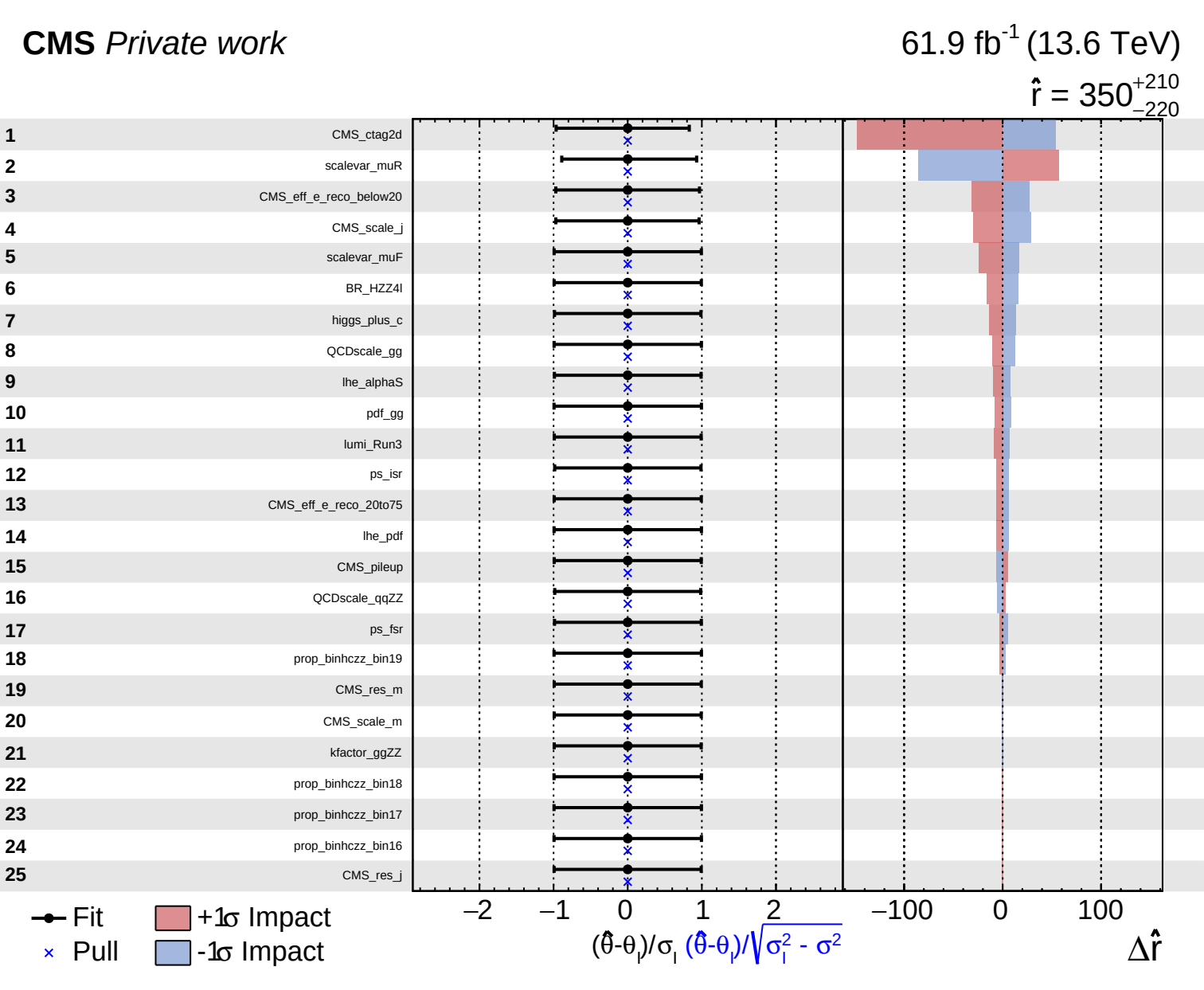
Quantile	r	kappa_c
2.5%	178.6641	33.63
16%	242.9138	44.79
50% (median)	**346.5000**	**62.75**
84%	510.8585	91.24
97.5%	730.3098	129.26

Stat only (--freezeParameters allConstrainedNuisances): r=273.0 / kappa_c=50.01.
Full syst: autoMCStats(10) does not move the no-Z+X median (346.5 with and without;
background MC-stat ~1.3% per bin vs ~50% data stat per bin).
kappa_c = sqrt(0.97 r + 0.03 r^2) (flat direction, F. Heyen thesis Eq. 2.29, B(H->cc)=0.03).

Systematics, no Z+X: type, value, and impact

Ranked by impact_r (combineTool.py -M Impacts, -t -1 --expectSignal 346.5, r in [0,1500], robustFit). r-hat = 346.5 +213.7/-220.7. MC stat = autoMCStats per-bin nuisances (prop_binhczz_bin<N>, N = MVA score bin 0-19).

#	Systematic	Type	Value (lnN size / pull)	Impact	dr (-1s / +1s)
1	**CMS_ctag2d**	**shape**	**pull +0.00 (-0.966/+0.830)**	**147.58**	**+53.6 / -147.6**
2	scalevar_muR	shape	pull +0.00 (-0.890/+0.930)	85.20	-85.2 / +57.3
3	CMS_eff_e_reco_below20	shape	pull +0.00 (-0.970/+0.969)	31.27	+26.7 / -31.3
4	CMS_scale_j	shape	pull +0.00 (-0.970/+0.963)	29.71	+28.7 / -29.7
5	scalevar_muF	shape	pull +0.00 (-0.993/+0.993)	23.67	+16.3 / -23.7
6	BR_HZZ4l	lnN	1.02 (Signal, Other_Higgs)	15.76	+15.8 / -15.6
7	higgs_plus_c	shape	pull +0.00 (-0.987/+0.988)	13.73	+13.1 / -13.7
8	QCDscale_gg	lnN	1.039 (Signal, ggZZ)	12.28	+12.3 / -10.0
9	lhe_alphaS	shape	pull +0.00 (-0.989/+0.990)	9.80	+7.7 / -9.8
10	pdf_gg	lnN	1.032 (Signal, ggZZ)	8.62	+8.6 / -8.2
11	lumi_Run3	lnN	1.014 (Signal, ggZZ, qqZZ, Other_Higgs)	8.48	+7.1 / -8.5
12	ps_isr	shape	pull +0.00 (-0.985/+0.989)	6.45	+5.7 / -6.4
13	CMS_eff_e_reco_20to75	shape	pull +0.00 (-0.992/+0.992)	6.31	+5.7 / -6.3
14	lhe_pdf	shape	pull +0.00 (-0.994/+0.994)	6.15	+6.1 / -6.0
15	CMS_pileup	shape	pull +0.00 (-0.992/+0.992)	6.05	-6.1 / +5.2
16	QCDscale_qqZZ	lnN	1.04 (qqZZ)	5.10	-5.1 / +3.0
17	ps_fsr	shape	pull +0.00 (-0.987/+0.993)	4.88	+4.9 / -3.1
18	prop_binhczz_bin19	MC stat	autoMCStats, pull (-0.994/+0.994)	3.20	+3.2 / -3.2
19	CMS_res_m	shape	pull +0.00 (-0.992/+0.992)	0.99	-1.0 / -0.8
20	CMS_scale_m	shape	pull +0.00 (-0.992/+0.992)	0.94	-0.9 / -0.8
21	kfactor_ggZZ	lnN	1.10 (ggZZ)	0.86	-0.9 / +0.8
22	prop_binhczz_bin18	MC stat	autoMCStats, pull (-0.994/+0.994)	0.72	+0.7 / -0.7
23	prop_binhczz_bin17	MC stat	autoMCStats, pull (-0.994/+0.994)	0.48	+0.5 / -0.5
24	prop_binhczz_bin16	MC stat	autoMCStats, pull (-0.994/+0.994)	0.33	+0.3 / -0.3
25	CMS_res_j	shape	pull +0.00 (-0.993/+0.991)	0.26	-0.2 / -0.3
26	prop_binhczz_bin15	MC stat	autoMCStats, pull (-0.994/+0.994)	0.22	+0.2 / -0.2
27	prop_binhczz_bin14	MC stat	autoMCStats, pull (-0.994/+0.994)	0.16	+0.2 / -0.2
28	prop_binhczz_bin13	MC stat	autoMCStats, pull (-0.994/+0.994)	0.12	+0.1 / -0.1
29	CMS_eff_e_reco_above75	shape	pull +0.00 (-0.994/+0.994)	0.10	+0.1 / -0.1
30	prop_binhczz_bin12	MC stat	autoMCStats, pull (-0.994/+0.994)	0.09	+0.1 / -0.1
31	prop_binhczz_bin11	MC stat	autoMCStats, pull (-0.994/+0.994)	0.06	+0.1 / -0.1
32	prop_binhczz_bin10	MC stat	autoMCStats, pull (-0.994/+0.994)	0.06	+0.1 / -0.1
33	prop_binhczz_bin9	MC stat	autoMCStats, pull (-0.994/+0.994)	0.04	+0.0 / -0.0
34	prop_binhczz_bin8	MC stat	autoMCStats, pull (-0.994/+0.994)	0.03	+0.0 / -0.0
35	CMS_eff_m_id	shape	pull +0.00 (-0.994/+0.994)	0.02	-0.0 / +0.0
36	prop_binhczz_bin7	MC stat	autoMCStats, pull (-0.994/+0.994)	0.01	+0.0 / -0.0
37	prop_binhczz_bin6	MC stat	autoMCStats, pull (-0.994/+0.994)	0.01	+0.0 / -0.0
38	prop_binhczz_bin5	MC stat	autoMCStats, pull (-0.994/+0.994)	0.01	+0.0 / -0.0
39	prop_binhczz_bin2	MC stat	autoMCStats, pull (-0.994/+0.994)	0.01	+0.0 / -0.0
40	prop_binhczz_bin4	MC stat	autoMCStats, pull (-0.994/+0.994)	0.01	+0.0 / -0.0
41	prop_binhczz_bin1	MC stat	autoMCStats, pull (-0.994/+0.994)	0.01	+0.0 / -0.0
42	prop_binhczz_bin3	MC stat	autoMCStats, pull (-0.994/+0.994)	0.00	-0.0 / -0.0
43	prop_binhczz_bin0	MC stat	autoMCStats, pull (-0.994/+0.994)	0.00	-0.0 / -0.0


CMS Private work
61.9 fb-1 (13.6 TeV)
r̂ = 350+210−220
1 CMS_ctag2d
2 scalevar_muR
3 CMS_eff_e_reco_below20
4 CMS_scale_j
5 scalevar_muF
6 BR_HZZ4l
7 higgs_plus_c
8 QCDscale_gg
9 lhe_alphaS
10 pdf_gg
11 lumi_Run3
12 ps_isr
13 CMS_eff_e_reco_20to75
14 lhe_pdf
15 CMS_pileup
16 QCDscale_qqZZ
17 ps_fsr
18 prop_binhczz_bin19
19 CMS_res_m
20 CMS_scale_m
21 kfactor_ggZZ
22 prop_binhczz_bin18
23 prop_binhczz_bin17
24 prop_binhczz_bin16
25 CMS_res_j
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
(θ̂-θ_I)/σ_I
−100
0
100
(θ̂-θ_I)/√(σ_I² - σ²)
Δr̂

HcZZ ctag2d -- Results with Z+X (2026-09-23)

Card: combine_run3_100150_ctag2d_rawsf_withzxCRmerged_noZXrate_jecshifts_automcstats_2026_09_23
Same MC/systematics as the no-Z+X card, plus Z+X from the fake-rate OS method (3P1F/2P2F)
on the hplusc_mva_4class_CR_merged production with pkatris's WZ-subtracted OS fake rates:
32.60 +- 4.2 events (per era 6.9/13.1/10.1/2.4), jet-inclusive (no c-jet requirement, by
decision). Z+X normalisation: ZX_norm 30% lnN only -- the free ZX_rate rateParam was removed
2026-09-23 (no Z+X control region in the fit; as in HIG-24-013). The datacard now applies the
m4l window to the Z+X input itself (the earlier '133' Z+X was pkatris's inclusive-m4l yield;
43.8 in the window).

Yields in [100,150] GeV

Process	Yield
Signal	0.0531
ggZZ	6.6779
qqZZ	36.3182
Other_Higgs	55.0776
Z+X	32.6047
Total bkg	**130.6784**

$S/\sqrt{B}$ = 0.0046

Expected limits (combine -M AsymptoticLimits -m 120 --run blind)

Quantile	r	kappa_c
2.5%	181.7578	34.17
16%	247.1201	45.52
50% (median)	**352.5000**	**63.79**
84%	522.5147	93.26
97.5%	749.9950	132.67

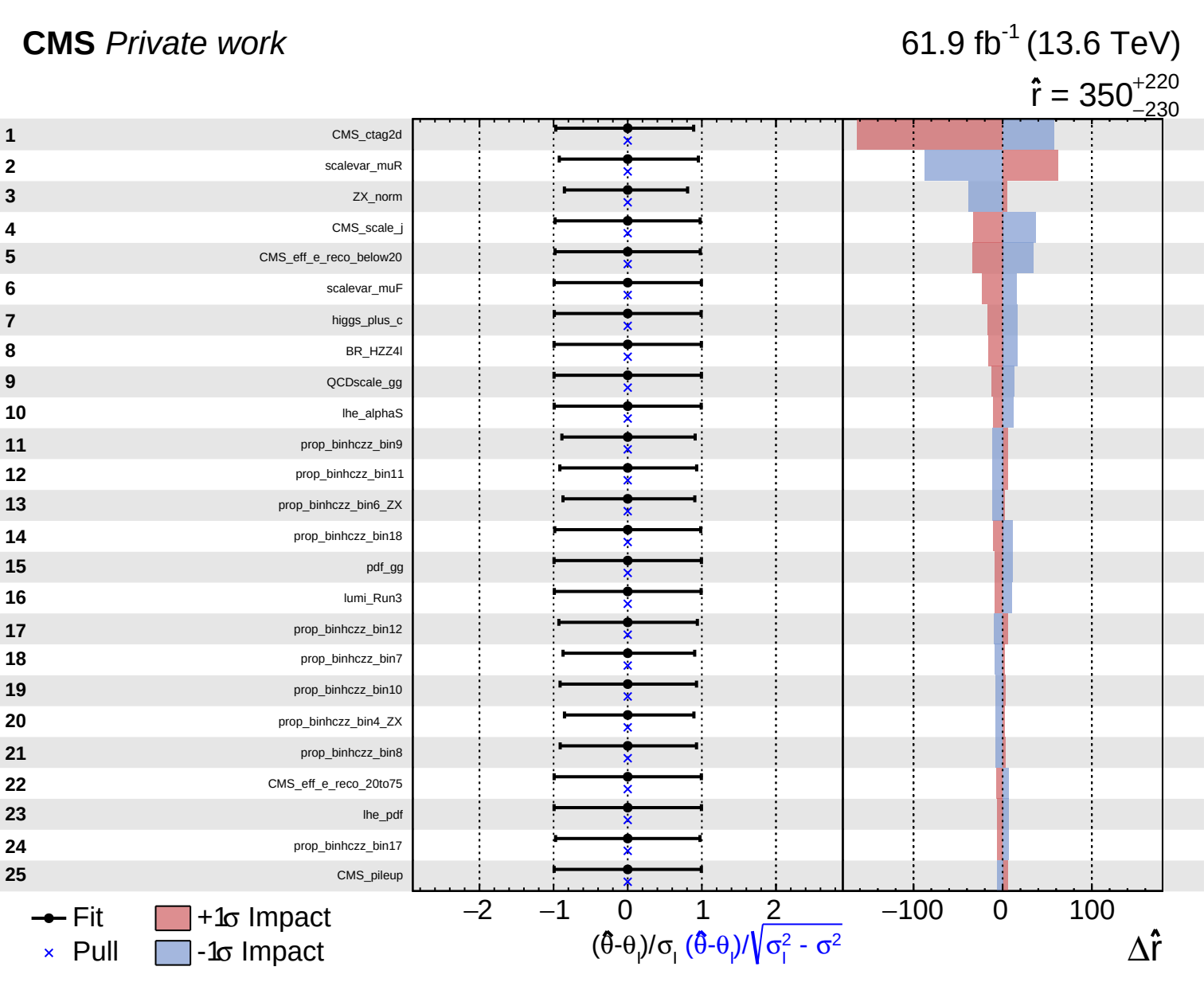
Comparison

Configuration	r (median)	kappa_c
no Z+X	346.5	62.75
with Z+X (free ZX_rate + ZX_norm, earlier today)	353.0	63.88
with Z+X (ZX_norm only, current)	352.5	63.79

Systematics with Z+X: type, value, and impact

Ranked by impact_r (-t -1 --expectSignal 352.5, r in [0,1500], robustFit). r-hat = 352.5 +218.3/-232.5.
ZX_norm is one-sided (-38/+5): inherited from the one-sided CMS_ctag2d (see last page).

#	Systematic	Type	Value (lnN size / pull)	Impact	dr (-1s / +1s)
1	**CMS_ctag2d**	**shape**	**pull +0.00 (-0.971/+0.890)**	**163.07**	**+57.5 / -163.1**
2	scalevar_muR	shape	pull +0.00 (-0.923/+0.953)	87.13	-87.1 / +62.3
3	ZX_norm	lnN	1.30 (ZX)	38.22	-38.2 / +5.0
4	CMS_scale_j	shape	pull +0.00 (-0.981/+0.977)	37.11	+37.1 / -33.1
5	CMS_eff_e_reco_below20	shape	pull +0.00 (-0.981/+0.979)	34.30	+34.3 / -34.0
6	scalevar_muF	shape	pull +0.00 (-0.993/+0.994)	22.78	+15.5 / -22.8
7	higgs_plus_c	shape	pull +0.00 (-0.990/+0.990)	16.87	+16.9 / -16.9
8	BR_HZZ4l	lnN	1.02 (Signal, Other_Higgs)	16.84	+16.8 / -16.1
9	QCDscale_gg	lnN	1.039 (Signal, ggZZ)	13.24	+13.2 / -12.5
10	lhe_alphaS	shape	pull +0.00 (-0.992/+0.992)	12.49	+12.5 / -10.3
11	prop_binhczz_bin9	MC stat	autoMCStats, pull (-0.888/+0.910)	11.78	-11.8 / +5.9
12	prop_binhczz_bin11	MC stat	autoMCStats, pull (-0.916/+0.930)	11.69	-11.7 / +5.5
13	prop_binhczz_bin6_ZX	MC stat	autoMCStats, pull (-0.873/+0.905)	11.66	-11.7 / +2.6
14	prop_binhczz_bin18	MC stat	autoMCStats, pull (-0.985/+0.985)	11.21	+11.2 / -10.8
15	pdf_gg	lnN	1.032 (Signal, ggZZ)	10.84	+10.8 / -8.6
16	lumi_Run3	lnN	1.014 (Signal, ggZZ, qqZZ, Other_Higgs, ZX)	10.66	+10.7 / -8.5
17	prop_binhczz_bin12	MC stat	autoMCStats, pull (-0.929/+0.940)	9.17	-9.2 / +5.5
18	prop_binhczz_bin7	MC stat	autoMCStats, pull (-0.872/+0.903)	8.53	-8.5 / +2.5
19	prop_binhczz_bin10	MC stat	autoMCStats, pull (-0.913/+0.928)	8.10	-8.1 / +2.8
20	prop_binhczz_bin4_ZX	MC stat	autoMCStats, pull (-0.853/+0.893)	7.86	-7.9 / +2.1
21	prop_binhczz_bin8	MC stat	autoMCStats, pull (-0.911/+0.928)	7.48	-7.5 / +2.7
22	CMS_eff_e_reco_20to75	shape	pull +0.00 (-0.993/+0.993)	6.76	+6.3 / -6.8
23	lhe_pdf	shape	pull +0.00 (-0.993/+0.994)	6.61	+6.6 / -6.4
24	prop_binhczz_bin17	MC stat	autoMCStats, pull (-0.973/+0.975)	6.51	+6.5 / -5.8
25	CMS_pileup	shape	pull +0.00 (-0.993/+0.993)	6.47	-6.5 / +5.7
26	prop_binhczz_bin5	MC stat	autoMCStats, pull (-0.870/+0.904)	6.33	-6.3 / +2.2
27	prop_binhczz_bin13	MC stat	autoMCStats, pull (-0.932/+0.942)	6.25	-6.3 / +1.6
28	ps_fsr	shape	pull +0.00 (-0.991/+0.994)	5.77	+5.8 / -4.0
29	prop_binhczz_bin3	MC stat	autoMCStats, pull (-0.830/+0.874)	5.55	-5.6 / +3.1
30	prop_binhczz_bin14	MC stat	autoMCStats, pull (-0.942/+0.949)	4.23	-4.2 / +0.6
31	prop_binhczz_bin16	MC stat	autoMCStats, pull (-0.948/+0.953)	3.46	+3.5 / -2.7
32	prop_binhczz_bin19	MC stat	autoMCStats, pull (-0.994/+0.994)	3.13	+3.1 / -3.1
33	prop_binhczz_bin1	MC stat	autoMCStats, pull (-0.898/+0.920)	2.53	-2.5 / +2.5
34	prop_binhczz_bin2_ZX	MC stat	autoMCStats, pull (-0.780/+0.844)	1.90	-1.7 / +1.9
35	ps_isr	shape	pull +0.00 (-0.990/+0.991)	1.52	+1.5 / -1.5
36	prop_binhczz_bin0	MC stat	autoMCStats, pull (-0.971/+0.974)	0.82	-0.8 / +0.8
37	QCDscale_qqZZ	lnN	1.04 (qqZZ)	0.78	+0.4 / -0.8
38	CMS_res_m	shape	pull +0.00 (-0.993/+0.993)	0.73	-0.7 / -0.3
39	prop_binhczz_bin15	MC stat	autoMCStats, pull (-0.951/+0.956)	0.63	+0.6 / -0.4
40	CMS_scale_m	shape	pull +0.00 (-0.993/+0.993)	0.57	-0.6 / -0.2
41	CMS_res_j	shape	pull +0.00 (-0.993/+0.992)	0.45	-0.3 / -0.4
42	kfactor_ggZZ	lnN	1.10 (ggZZ)	0.16	-0.2 / +0.1
43	CMS_eff_e_reco_above75	shape	pull +0.00 (-0.994/+0.994)	0.06	+0.1 / -0.1
44	CMS_eff_m_id	shape	pull +0.00 (-0.994/+0.994)	0.02	-0.0 / +0.0
45	prop_binhczz_bin6_qqZZ	MC stat	autoMCStats, pull (-0.994/+0.994)	0.02	+0.0 / -0.0
46	prop_binhczz_bin4_qqZZ	MC stat	autoMCStats, pull (-0.994/+0.994)	0.01	+0.0 / -0.0
47	prop_binhczz_bin2_qqZZ	MC stat	autoMCStats, pull (-0.994/+0.994)	0.01	+0.0 / -0.0
48	prop_binhczz_bin6_ggZZ	MC stat	autoMCStats, pull (-0.994/+0.994)	0.00	+0.0 / -0.0
49	prop_binhczz_bin6_Signal	MC stat	autoMCStats, Poisson (n_eff 9)	0.00	+0.0 / -0.0
50	prop_binhczz_bin6_Other_Higgs	MC stat	autoMCStats, pull (-0.994/+0.994)	0.00	+0.0 / -0.0
51	prop_binhczz_bin2_ggZZ	MC stat	autoMCStats, pull (-0.994/+0.994)	0.00	+0.0 / -0.0
52	prop_binhczz_bin4_ggZZ	MC stat	autoMCStats, pull (-0.994/+0.994)	0.00	+0.0 / -0.0
53	prop_binhczz_bin4_Signal	MC stat	autoMCStats, Poisson (n_eff 5)	0.00	+0.0 / -0.0
54	prop_binhczz_bin2_Signal	MC stat	autoMCStats, Poisson (n_eff 6)	0.00	+0.0 / -0.0
55	prop_binhczz_bin4_Other_Higgs	MC stat	autoMCStats, pull (-0.994/+0.994)	0.00	+0.0 / -0.0
56	prop_binhczz_bin2_Other_Higgs	MC stat	autoMCStats, pull (-0.994/+0.994)	0.00	+0.0 / -0.0


CMS Private work
61.9 fb-1 (13.6 TeV)
r̂ = 350+220−230
1 CMS_ctag2d
2 scalevar_muR
3 ZX_norm
4 CMS_scale_j
5 CMS_eff_e_reco_below20
6 scalevar_muF
7 higgs_plus_c
8 BR_HZZ4l
9 QCDscale_gg
10 lhe_alphaS
11 prop_binhczz_bin9
12 prop_binhczz_bin11
13 prop_binhczz_bin6_ZX
14 prop_binhczz_bin18
15 pdf_gg
16 lumi_Run3
17 prop_binhczz_bin12
18 prop_binhczz_bin7
19 prop_binhczz_bin10
20 prop_binhczz_bin4_ZX
21 prop_binhczz_bin8
22 CMS_eff_e_reco_20to75
23 lhe_pdf
24 prop_binhczz_bin17
25 CMS_pileup
−2 −1 0 1 2
−100 0 100
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/σ0 (θ̂-θ0)/√σ0² - σ²
Δr̂

Stat-only limits: with vs. without Z+X

combine -M AsymptoticLimits --freezeParameters allConstrainedNuisances on the current cards
(all systematics, ZX_norm and the autoMCStats parameters frozen; no free parameter other than r).

Quantile	r (no ZX)	kc (no ZX)	r (with ZX)	kc (with ZX)	dr	dr %
2.5%	140.7656	27.04	144.6328	27.71	+3.87	+2.7%
16%	191.3866	35.84	195.5830	36.57	+4.20	+2.2%
50% (median)	**273.0000**	**50.01**	**280.5000**	**51.31**	**+7.50**	**+2.7%**
84%	396.5095	71.42	407.9617	73.41	+11.45	+2.9%
97.5%	553.1971	98.58	570.0557	101.50	+16.86	+3.0%

Stat only, Z+X loosens the median by 2.7% (273.0 -> 280.5);
with full systematics by 1.7% (346.5 -> 352.5).
Systematics loosen the median by 27% (no Z+X) / 26% (with Z+X).

What drives the Z+X impact? (with-Z+X card)

Impacts from Asimov fits with r injected at each variant's median; dr for -1 sigma / +1 sigma.

Variant	median r	ZX_norm impact	CMS_ctag2d impact	other Z+X NP
Free ZX_rate + ZX_norm (before 09-23)	353.0	-0.9 / +0.1	+57 / -163	ZX_rate -82.9 / +4.9
A: production (ZX_norm only)	352.5	-38.2 / +5.0	+57 / -163	--
B: CMS_ctag2d frozen	337.5	-3.6 / -1.7	frozen	--
C: CMS_ctag2d Down = mirror of Up	528.0	-46.9 / +41.0	+298 / -208	--

B: without CMS_ctag2d the ZX_norm impact nearly vanishes -- its size and asymmetry come from the fit compensating a Z+X shift through CMS_ctag2d. A ZX_norm scan with ctag2d frozen is a shallow hump (r-hat 346.9 ... 352.5 ... 349.0 over +-1.1 sigma), hence the same-sign -3.6/-1.7.

C: a symmetric CMS_ctag2d makes ZX_norm symmetric, so the asymmetry is inherited from ctag2d (Up: Signal +36.5%, Other_Higgs +15.3%, qqZZ +8.6%; Down: -3.1%, -4.4%, +0.3%). C is a diagnostic only, not a valid model.

Mechanism: Z+X down -> the fit refills the low-score bins via ctag2d Up (Other_Higgs +29% in bins 0-9), which also raises the signal yield -> r-hat down. Z+X up needs ctag2d Down, which is ~flat -> other NPs absorb it -> r-hat ~unchanged.

Rebinning (3-20 bins, merged) does not change the ZX_norm or CMS_ctag2d impacts.
Open: the one-sided CMS_ctag2d Up (L0 up/down, with the ctag2d calibration contact).